Xiang Pan

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Research Interests

My research lies at the intersection of human–robot interaction and embodied AI. My previous work has focused on developing expressive manipulation strategies to enhance the transparency, legibility, and naturalness of robot behavior in human-shared environments.
Currently, I investigate how robots can infer and respond to human intentions by leveraging large language models (LLMs) and vision language models (VLMs), informed by insights from psychology to enable socially appropriate behavior in human–robot interaction.

Employment

02/2026 – Present
Kyoto University – Kyoto, Japan
Postdoctoral Researcher
Advisor: Prof. Takayuki Kanda

Education

04/2022 – 01/2026
Kyoto University – Kyoto, Japan
Ph.D. in Social Informatics
Advisor: Prof. Takayuki Kanda

10/2020 – 08/2021
Northeast Normal University – Changchun, China
Japanese Language Studies

09/2017 – 06/2020
Zhejiang University – Hangzhou, China
M.S. in Instrument and Meter Engineering
Advisor: Prof. Hong Zhou

09/2013 – 07/2017
Anhui University of Technology – Ma'anshan, China
B.S. in Electronic and Information Engineering
Advisor: Prof. Bing Wang

Full Papers in Conference Proceedings

2026
Communicating Object Relations through Robot Gestures.
Xiang Pan*, Malcolm Doering, Takayuki Kanda. In Proceedings of ACM/IEEE International Conference on Human Robot Interaction 2026.
[CORE A*, Top-tier HRI venue, 23.2% acceptance rate (129/557)]

2025
Communicating Physical Properties through Robot Object Manipulation.
Xiang Pan*, Malcolm Doering, Stela H. Seo, Takayuki Kanda. In Proceedings of ACM/IEEE International Conference on Human Robot Interaction 2025.
[CORE A*, Top-tier HRI venue, 25% acceptance rate (100/400)]

2024
What Is Your Other Hand Doing, Robot? A Model of Behavior for Shopkeeper Robot's Idle Hand.
Xiang Pan*, Malcolm Doering, Takayuki Kanda. In Proceedings of ACM/IEEE International Conference on Human Robot Interaction 2024.
[CORE A*, Top-tier HRI venue, 24.7% acceptance rate (87/352)]

2019	**Efficient Barcode Localization Method for Low-Quality Images.** Xiang Pan, Dong Li, Weijia Wu, Hong Zhou *. In Proceedings of International Conference on Graphics and Signal Processing 2019.

Invention Patents

2024	**High Voltage LED Chip Set, LED Light Source for Plant Light Supplementation and Illuminating Device.** Xiang Pan, Xuke Li. US Patent (US12051677B2), Granted
2023	**LED Light Source for Plant Light Supplementation and Lamp Comprising the Same.** Xuke Li, Xiang Pan. European Patent (EP3767167B1), Granted
2022	**LED Light Source for Supplemental Lighting for Plants and Lamp with Light Source.** Xuke Li, Xiang Pan. US Patent (US11419278B2), Granted

Teaching Experience

2025	**Information System Analysis** Teaching Assistant

Academic Service

Conference Workshop Organisor

Multi-Agentic Systems in HRI (MAgicS-HRI): Bridging Design and Real-World Challenges for End Users, HRI' 26

Journal Review

IEEE Robotics and Automation Letters (RA-L)

ACM Transactions on Human-Robot Interaction (THRI)

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)

Conference Review

2023	ACM Conference on Human Factors in Computing Systems (CHI)
2025, 2026	ACM/IEEE International Conference on Human-Robot Interaction (HRI)
2025, 2026	ACM Conference on Designing Interactive Systems (DIS)
2025, 2026	ACM Interaction Design and Children Conference (IDC)
2025	IFIP TC13 International Conference on Human-Computer Interaction (INTERACT)
2025	IEEE World Haptics Conference (WHC)
2026	ACM Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutoUI)
2024	IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)

2024 International Conference on Human-Agent Interaction (HAI)

Student Volunteer

2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Invited Talks

2025 **Communicating Object Relations through Robot Gestures.**
Invited Speaker, Center for Digital Media Computation, Xiamen University

2024 **Communicating Physical Properties through Robot Object Manipulation.**
Invited Speaker, Center for Digital Media Computation, Xiamen University

Selected Awards

2021 **Japanese Government (MEXT) Scholarship**
Ministry of Education, Culture, Sports, Science and Technology

2020 **Excellent Postgraduate Student Award**
Zhejiang University

2018 **Outstanding Graduate Student Leader Award**
Zhejiang University

2017 **Outstanding Graduate Student Award**
Anhui Province

2017 **Youth May Fourth Medal**
Anhui University of Technology (Highest Undergraduate Honor, 10 Awardees University-wide)

Robot Platforms

TIAGo++; Robovie; Pepper